SecureHire: An Explainable Hybrid Framework Combining NLP, Rule-Based Red-Flag Analysis and Government Verification for Fake Job Posting Detection

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ABSTRACT

Online recruitment platforms are increasingly exploited by fraudsters who publish fake job advertisements to harvest personal data, application fees or unpaid work. Existing detectors mostly analyse text alone, behave as black boxes, are trained on a single English-only imbalanced dataset, and never verify the recruiting company. This paper presents SecureHire, a seven-layer hybrid framework that combines TF-IDF, Bag-of-Words and contextual embeddings with Random Forest, ANN, mBERT and XLM-RoBERTa classifiers; rule-based red-flag checks (advance fees, sensitive-document requests, unrealistic pay, free e-mail domains, WhatsApp-only contact, urgency, no-interview promises); government and external verification of company identity (MCA, GST, Companies House, SEC EDGAR, phone and domain checks); hierarchical clustering to expose scam sub-types; and SHAP/LIME explanations. The output is a Real / Medium-Risk / Fake verdict, a risk score and apply-or-avoid guidance. We also implemented TrueHire, a browser-only prototype of the rule and verification layers supporting English, Hindi and Hinglish, with GSTIN checksum validation, recruiter-e-mail and domain checks, duplicate-posting detection and a user-feedback dashboard. On its three built-in demonstration postings the prototype assigned risk scores of 3 (genuine corporate role), 100 and 98 (two scam postings). The learned-model evaluation is still pending; this paper therefore reports the design, the prototype behaviour and a comparison with accuracies reported in the literature, and defines the evaluation protocol for the full system.

Index Terms — fake job posting detection, online recruitment fraud, explainable AI, multilingual transformers, government verification

I. INTRODUCTION

The internet has made job hunting faster, but it has also given fraudsters an easy channel to reach job seekers. Fake postings imitate genuine openings to collect personal data, application fees or unpaid work. The Federal Trade Commission reported that Americans lost about US $450 million to fake job scams in 2024, Australia reported a total loss of AU $2.74 billion with job-scam losses up 150% over 2022, and in the EMSCAD dataset 12% of the sampled Workable postings were fraudulent. Automated detection is needed because manual review cannot scale to thousands of daily postings, scams target students, entry-level and flexible-work seekers, and fraudulent postings mimic genuine ones closely enough to defeat simple keyword filters.

A. Related Work

Early work framed online recruitment fraud as text classification. Vidros et al. released the EMSCAD dataset and used Random Forest with Bag-of-Words features on linguistic and structural patterns [1]. Alghamdi and Alharby combined SVM and Random Forest with company-profile, logo, industry and job-detail features [2]. Classical learners such as SVMs [3], decision trees [4] and Random Forests [5] remain strong baselines on TF-IDF and structured features. Because fake postings are a small minority in EMSCAD, imbalance handling such as SMOTE [6] improves classification. Deep models, including Bi-LSTM [7], sequence networks with GloVe embeddings and CNNs, outperform several traditional methods at higher compute cost. Transformer encoders such as BERT [8] and cross-lingual XLM-RoBERTa [9] have more recently been applied to recruitment fraud, for example Fraud-BERT [10], and the related problem of ghost jobs has also been studied [11]. A multilingual explainable-AI study reports 97.8% accuracy with XLM-RoBERTa, and the hybrid JobShield system reports 87.4% accuracy (F1 = 0.86) by combining TF-IDF, Random Forest, rule checks and external API verification.

B. Research Gaps

The surveyed systems share five limitations: (1) they act as black boxes and give no reason for flagging a posting; (2) they analyse text only and do not verify recruiter identity or company legitimacy; (3) most are trained on a single, imbalanced, English-only dataset; (4) they struggle with multilingual and code-mixed postings in regional languages; and (5) high-accuracy deep and transformer models need heavy compute and frequent retraining.

C. Contributions

This paper makes the following contributions:

1. SecureHire, a unified seven-layer architecture merging EDA/clustering, classical and transformer classifiers, rule-based checks, government/external verification and SHAP/LIME explainability.
2. A government-verification step that cross-checks company registration, tax and domain details with official sources (MCA, GST, Companies House, SEC

EDGAR) and reports a Verified / Not Verified status with every prediction.

3. TrueHire, a working browser-based prototype of the rule and verification layers for English, Hindi and Hinglish, with a transparent weighted scoring scheme, GSTIN checksum validation and duplicate-posting detection.
4. A defined evaluation protocol (80/20 split; accuracy, precision, recall, F1 and confusion matrix) and an honest comparison with accuracies reported in prior work.

II. PROPOSED WORK: THE SECUREHIRE FRAMEWORK

SecureHire is organised as seven layers followed by the final fraud-detection result: Data, Preprocessing, Feature Extraction, Machine Learning, Rule-Based and Verification, Explainability and Evaluation, and Recommendation and UI.

A. Data Layer and Collection

Multilingual postings are gathered from public datasets (EMSCAD, Kaggle Balanced Fake Job Postings) and from live recruitment platforms and APIs. Web scraping collects company website and domain details, and real-time listings are monitored for new postings.

B. Preprocessing

Incomplete fields are handled first. HTML tags, URLs, punctuation and duplicate postings are removed; text is lowercased and normalised, tokenised with NLTK or spaCy, stripped of stop-words and lemmatised. The language of each posting is detected so that multilingual and code-mixed postings are routed to suitable models.

C. Feature Extraction

Cleaned text is converted into TF-IDF vectors or Bag-of-Words features for classical models and ANN, and into contextual transformer embeddings for mBERT and XLM-RoBERTa [8], [9]. Dense word representations in the style of [12] may be used for sequence models.

D. Machine Learning Layer

Random Forest [5], an Artificial Neural Network, mBERT and XLM-RoBERTa classify each posting as Real or Fake. Models are trained and compared on the same 80/20 train/test split, with SMOTE [6] considered for class imbalance. Hierarchical clustering groups postings by theme and exposes scam sub-types (for example a "client relations" scam) and geographic hotspots; Tableau and Orange dashboards (pie, bar, choropleth, word cloud) support the exploratory analysis.

E. Rule-Based and Verification Layer

This layer adds precision beyond text statistics. Salary, e-mail and keyword rules flag free e-mail domains, WhatsApp-only contact, unrealistic salary and urgency phrases. The government business-registration check cross-checks company details (registration, GST, domain, licence) against MCA, GST and UIDAI in India, Companies House in the UK and SEC EDGAR in the US, and returns Verified / Not Verified. External APIs (Twilio, WhoisXML, OpenCage, OpenCorporates) check phone numbers and domains. Planned features include a Behavioral Tracker for interview-farming patterns, an IP and Assignment Guard for suspicious unpaid-work requests, and Auto Quarantine of high-risk postings.

F. Explainability and Evaluation Layer

SHAP [13] provides global feature importance and LIME [14] explains individual predictions by highlighting the words behind the verdict, such as "registration fee" or "urgent hiring". Performance is measured by accuracy, precision, recall, F1-score (appropriate because fake postings form a minority class) and the confusion matrix.

G. Recommendation and UI Layer

A Streamlit/Flask web application shows the Real / Medium-Risk / Fake verdict, risk score, government-verification status, suspicious features found, and apply-or-avoid guidance.

H. TrueHire Prototype Implementation

TrueHire is a single HTML/JavaScript page that runs entirely in the browser, so no posting is uploaded. It implements the rule and verification layers as follows.

Field extraction. Regular expressions extract company, job title, location, recruiter e-mail, domain and an Indian GSTIN from the pasted text.

Rule engine. Eleven red-flag rules fire on English, Devanagari Hindi and romanised Hinglish patterns. Each rule has a weight (Table I). The scam probability is the sum of the weights of fired rules, capped at 98%; if no rule fires it is fixed at 3%.

TABLE I. TrueHire red-flag rules and weights

Rule	Weight
Advance registration or security fee requested	60
PAN, Aadhaar or bank details requested upfront	25
Task-style work (likes, reviews, copy-paste)	25
Unrealistically high pay for unskilled work	20
Pushes chat on WhatsApp or Telegram	12
Selection promised without interview	12
Guaranteed income or placement	12
Overseas job with visa or processing money	10
Free personal e-mail address	8
Artificial urgency (limited seats)	8
Company name missing or vague	8

Verification checks. Six checks are shown to the user: (i) AI fraud analysis from the rule engine; (ii) government GSTIN verification, which validates the 15-character format, state code and checksum digit and fails invalid numbers (+20 risk), but reports a valid format only as "caution" because the registered name cannot be confirmed offline; (iii) company website and domain, which adds +6 risk for throwaway top-level domains (.xyz, .top, .click, etc.); (iv) recruiter e-mail, which adds +6 risk for free providers; (v) salary and offer anomaly; and (vi) recycled-posting detection, which fingerprints the posting text and adds +12 risk when identical text was previously seen with different contact details, a typical sign of a copied genuine-looking advert.

Final score. The final risk score is the scam probability plus the check risks, clamped to 0–100. Scores of 60 or more are High, 30–59 Medium and below 30 Low, mapped to "Do not apply", "Apply with caution" and "Looks safe to apply".

Explainability and usability. The report lists every fired red flag and a plain-language summary that can be switched between English, Hindi and Hinglish. Users can export a PDF, copy a summary, generate a cyber-crime report summary for the National Cyber Crime Reporting Portal (helpline 1930) for medium and high-risk results, and mark a result as wrong. Corrections are stored locally and feed a dashboard of scanned postings.

III. RESULTS AND DISCUSSION

A. Prototype Behaviour

The TrueHire rule engine was run on its three built-in demonstration postings (Table II). The genuine Tata Consultancy Services posting, which names its careers portal and states that no fee is charged, triggers no rule and receives the minimum score. The work-from-home fee posting triggers five rules and the Hinglish posting six, both landing in the High band. The Hinglish posting shows the multilingual rules working on romanised Hindi ("bina interview", "jaldi apply", "registration fees").

TABLE II. TrueHire output on the built-in demonstration postings

Posting	Rules fired	Risk score	Verdict
Legitimate corporate IT role (TCS)	none	3	Low – looks safe to apply
Work-from-home fee scam	advance fee, sensitive documents, high pay, free e-mail, urgency	100	High – do not apply
Hinglish suspicious post	advance fee, high pay, WhatsApp/Telegram, no interview, urgency, vague company	98	High – do not apply

For the genuine posting the GSTIN and domain checks return "could not verify" because no GSTIN is present and domain age cannot be queried inside a browser, while the e-mail check passes on the company domain tcs.com. This illustrates a deliberate design choice: the prototype reports "could not verify" instead of silently passing, and the full system closes this gap with live registry and WHOIS lookups. These three samples demonstrate behaviour only; they are not a statistical evaluation, and the rules were written with these scam patterns in mind.

B. Comparison with Accuracies Reported in the Literature

Table III collects the accuracies reported by prior studies reviewed for this work. They come from different datasets, splits and settings and are therefore indicative rather than directly comparable.

TABLE III. Accuracies reported in the reviewed literature

Approach	Technique	Reported accuracy
Naive Bayes / SVM / Logistic Regression	Classical ML on TF-IDF	80–89%
Random Forest / AdaBoost / Gradient Boosting	Tuned ensembles	Above 98% (reported)
Sequential NN + GloVe, Bi-LSTM, GRU / Attention-GRU	Deep learning	Above 97% (reported)
mBERT / XLM-RoBERTa (explainable multilingual study)	Multilingual transformer	97.8% (XLM-RoBERTa, best)
JobShield	TF-IDF + Random Forest + rules + API checks	87.4% (F1 = 0.86); 85–90%

Pure text models reach very high accuracy on EMSCAD, but they are English-only, unexplained and unverified against real company records. The hybrid JobShield system trades some headline accuracy for rule and API evidence. SecureHire aims to combine the accuracy of transformer models with that evidence, and adds government registration checks and explanations that none of the compared systems provide together (Table IV).

TABLE IV. Capability comparison

Aspect	Existing systems	SecureHire
Analysis basis	Text only	Text + rules + government/external verification
Explainability	Black box	SHAP and LIME
Language support	Single-language, English-only	Multilingual and code-mixed (mBERT, XLM-RoBERTa)
Company legitimacy	Not verified	MCA / GST / UIDAI, Companies House, SEC EDGAR
Data	Single imbalanced dataset	Public datasets plus live postings
Output	Real / Fake label	Real / Medium-Risk / Fake, risk score, apply-or-avoid guidance

C. Evaluation Protocol and Status

Random Forest, ANN, mBERT and XLM-RoBERTa will be trained and compared on the same 80/20 split using accuracy, precision, recall, F1-score and the confusion matrix, and SHAP/LIME outputs will be inspected to confirm that the words driving verdicts are sensible. Because the work is at the design and prototype stage, these trained-model results have not yet been recorded and Table V is reserved for them.

TABLE V. Trained-model results (to be completed)

Model	Accuracy	Precision	Recall	F1
Random Forest	–	–	–	–
ANN	–	–	–	–
mBERT	–	–	–	–
XLM-RoBERTa	–	–	–	–

D. Limitations

The work is at the design stage with no recorded experimental results for the learned models. Public data is mostly English-only and imbalanced, so multilingual and code-mixed data must still be collected and labelled.

Government and external verification depends on API availability, access terms and internet connectivity; the browser prototype cannot confirm registered names or domain age. Transformer models need a CUDA-capable GPU and periodic retraining. Rule-based flags can be evaded by scammers who learn the patterns, which is why they are combined with learned models and verification.

IV. CONCLUSION AND FUTURE WORK

This paper presented SecureHire, a hybrid fake job posting detection framework that addresses the black-box, text-only and English-only weaknesses of existing systems by combining NLP and machine-learning classifiers, rule-based red-flag analysis, government and external company verification and SHAP/LIME explanations, and delivering a verdict, risk score and apply-or-avoid guidance. The TrueHire prototype demonstrates the rule and verification layers for English, Hindi and Hinglish with transparent, weighted and explainable scoring. Future work includes training and benchmarking the four classifiers on EMSCAD plus collected multilingual data; extending coverage to more regional languages and code-mixed text; integrating live MCA/GST, Companies House and SEC EDGAR lookups; strengthening the Behavioral Tracker and IP and Assignment Guard; tracking new scam sub-types and hotspots through clustering dashboards; and deploying browser-extension, mobile and placement-cell versions with continuous learning from user feedback.

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